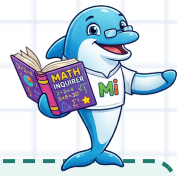
COPY ON THE GRID!

Name _____ Date _____

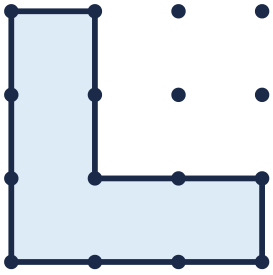


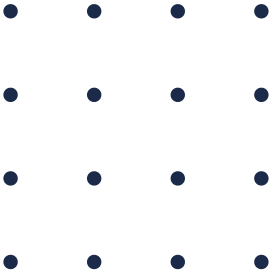
After question 3 you may choose: carry on, or try **Think harder**.

Count the steps between the dots, not the dots. Say it out loud, then draw or write it.

No.S1

Mei draws an L shape on a grid. Copy it on the empty grid.

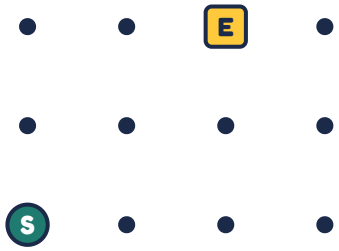




The L shape is ☐ steps across and ☐ steps up.

No.S2

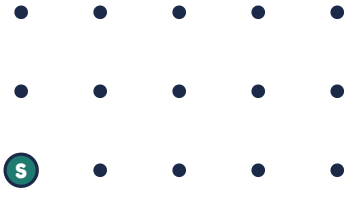
A counter starts at S. It moves only right and up to reach E.



The counter moves ☐ right and ☐ up.

No.S3

Start at S. Go 4 steps right. Then go 1 step up.



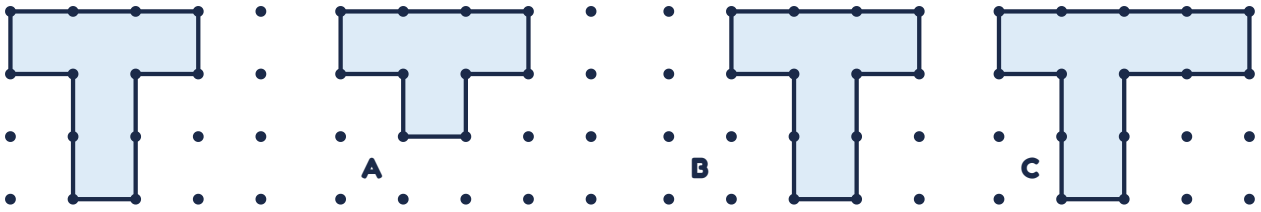
Draw the path. Then draw a ring round the dot where you stop.

Name _____ Date _____

Use your grid page and a button. A ruler or the edge of a card helps with straight lines.

No.S4

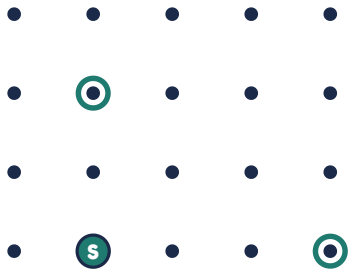
Look at the first picture. Which picture is an exact copy of it?



Circle A, B or C. **A** **B** **C**

No.S5

S and the two ringed dots are three corners of a rectangle.



(a) Draw the missing corner. Join the dots to finish the rectangle.

(b) The missing corner is ☐ right and ☐ up from S.

No.S6

Look at the row of dots. Tick True or False for each sentence.

Statement	True	False
There are 5 dots in the row.	☐	☐
There are 5 steps from the first dot to the last dot.	☐	☐
There are 4 steps from the first dot to the last dot.	☐	☐

Put a tick (✓) in the correct box.

Name _____ Date _____

Think about every route. List them in order so none is missed. Choose any question, in any order.

No.T1

A counter goes from S to E. It moves only right or up. The grid is 3 steps across and 1 step up.

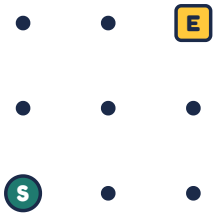


(a) One route is R R R U. R means right and U means up. Write the other routes: _____

(b) There are ☐ routes in all.

No.T2

A counter goes from S to E. It moves only right or up. The grid is 2 steps across and 2 steps up.

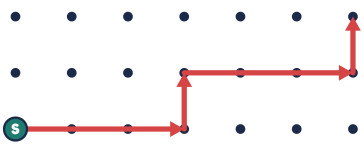


(a) Three routes start with U: U U R R, U R U R and U R R U. Write the routes that start with R: _____

(b) There are ☐ routes in all.

No.T3

Ravi's counter walks 3 right and 1 up, again and again. The picture shows 2 sets.



(a) After 3 sets, the counter is ☐ right and ☐ up from S.

(b) After 5 sets, the counter is ☐☐ right and ☐ up from S.

Name _____ Date _____

Look at the boxes that are filled in and see how they are made. Choose any question. You do not have to finish them all.

No.G1

The boxes show how many routes reach each corner. S is the bottom left box. E is the top right box.

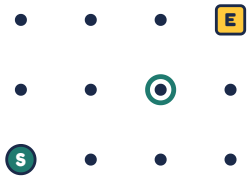
1	☐	☐	☐
1	2	3	☐
1	1	1	1

(a) Fill in the empty boxes.

(b) There are ☐☐ routes from S to E.

No.G2

A counter goes from S to E, moving only right or up. The grid is 3 steps across and 2 steps up. The route must pass the ringed corner.



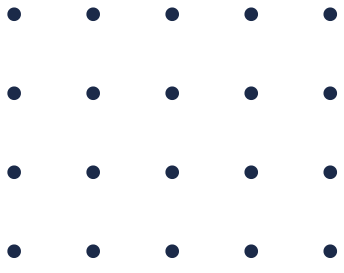
(a) Routes from S to the ringed corner: ☐

(b) Routes from the ringed corner to E: ☐

(c) Routes from S to E that pass the ringed corner: ☐

No.G3

Draw a shape with 4 corners on the grid. A is its bottom left corner. Many answers are right.

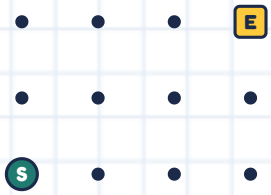


B is ☐ right and ☐ up from A. C is ☐ right and ☐ up from A. D is ☐ right and ☐ up from A.

Name _____ Date _____

Investigation: Same steps, different routes

First: Draw two different routes from S to E. Use a different line for each. Count the steps in each route: _____ steps and _____ steps.



Next: Do all the routes from S to E have the same number of steps? Write why.

Exit ticket

On your own, at the end of the lesson. No looking back.

No.E1

A counter starts at S and moves only right and up to reach E.



The counter moves ☐ right and ☐ up.

No.E2

A counter moves 2 right and 3 up. Then it moves 3 right and 1 up.

In all it moves ☐ right and ☐ up.

No.E3

A figure is 5 steps across and 2 steps up. A child draws a copy that is 6 steps across and 2 steps up.

(a) Is it an exact copy? Circle Yes or No. **Yes** **No**

(b) An exact copy is ☐ steps across.

Look-back There are no right answers. You can pass. Nobody reads it out.

Something I did well: .....

The hardest part was …
- ☐ counting steps, not dots
- ☐ copying exactly
- ☐ counting the routes
- ☐ something else

What helped me:
- ☐ the grid page
- ☐ my button
- ☐ drawing the routes
- ☐ I was not stuck

Next time I will try:
- ☐ counting steps
- ☐ saying where to go
- ☐ finding every route
- ☐ asking a question

☐ I am still stuck (my grown-up will photograph this page for my tutor)

☐ I would like to pass

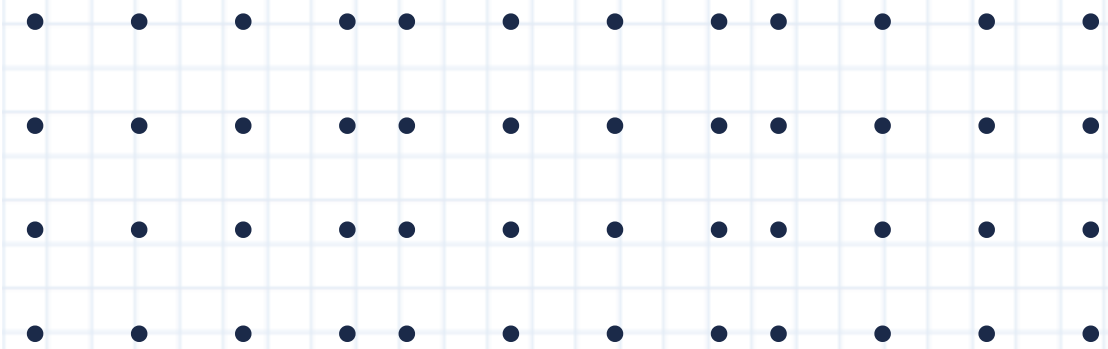
Name _____ Date _____

Kit page: grid page

No cutting: this page is for drawing on. The three small grids are route grids for walking a button. The two wide grids are copy grids: the dashed line splits each one, so copy the figure from the screen into the right-hand half. No printer? Draw rows of dots on paper.

No cutting needed: keep this page flat to use in the lesson. Print it on its own sheet, or on one side only.

Grids: keep this page flat to draw on. No cutting.



Copy grids: draw the other half

